

Coagulation — Hemophilia: Inhibitors, Efficizumab & Rare Factors



1. Inhibitor = alloantibody

WHAT YOU SEE

Bottom-left 'inhibitor = alloantibody neutralizes infused FVIII'

WHAT IT ENCODES

An inhibitor is a neutralizing IgG ALLOantibody that the patient develops against infused (foreign) factor — anti-FVIII in hemophilia A, anti-FIX in B. It is the most serious treatment complication of hemophilia, because standard factor replacement stops working: the antibody binds and inactivates the infused factor. Inhibitors are quantified in Bethesda units (BU).

BOARD TRAP

A previously well-controlled hemophiliac who suddenly stops responding to adequate factor doses has an INHIBITOR, not a dosing error or product failure — the next step is a mixing study and a Bethesda titer, not simply more factor.

2. Severe A, highest inhibitor risk

WHAT YOU SEE

'Wanted' poster severe hemophilia A, early exposure, African descent, family history

WHAT IT ENCODES

Inhibitor risk is highest in SEVERE hemophilia A and peaks during the first ~10–20 cumulative exposure days (typically early childhood). Additional risk factors: large gene deletions/null mutations (no endogenous protein for tolerance), family history of inhibitors, African ancestry, and intensive factor exposure (e.g., around surgery). Overall ~25–30% of severe hemophilia A patients develop them.

BOARD TRAP

Inhibitors are far MORE common in hemophilia A than B (~25–30% vs ~3–5%), and far more common in severe than mild disease — so the high-risk patient is the severe-A child in his first exposure days, not the mild adult.

3. Mixing study won't correct

WHAT YOU SEE

A center diagnostic panel '~60% high responder' with the inhibitor-titer concept — the aPTT that fails to correct on mixing and worsens on incubation

WHAT IT ENCODES

In a deficiency the aPTT corrects on a 1:1 mix with normal plasma; with an INHIBITOR the aPTT fails to correct (and a time/temperature-dependent FVIII inhibitor corrects initially then prolongs after incubation at 37°C). Titer is reported in Bethesda units: low responder <5 BU, high responder ≥5 BU (~60% are high responders). This distinction drives treatment.

BOARD TRAP

FAILURE TO CORRECT on the mixing study = inhibitor, not deficiency — and the FVIII autoantibody pattern is classically TIME- and TEMPERATURE-dependent (incubated mix prolongs); a simple non-correcting mix can also be a lupus anticoagulant, so confirm with the Bethesda assay.

4. Bypassing agents (aPCC, rFVIIa)

WHAT YOU SEE

Top-center 'BYPASS SHELF' stocked with activated PCC and rFVIIa vials tagged 'bypasses intrinsic' — agents that act downstream of the inhibited factor

WHAT IT ENCODES

Acute bleeds in patients with HIGH-titer inhibitors are treated with BYPASSING agents that work around the inhibited factor: activated prothrombin complex concentrate (aPCC = FEIBA) and recombinant activated factor VIIa (rFVIIa). They activate the common pathway downstream of FVIII/FIX, restoring thrombin generation regardless of the antibody. Low-titer inhibitors may still be overwhelmed with high-dose factor.

BOARD TRAP

Simply giving MORE FVIII does not work against a high-titer inhibitor — the antibody neutralizes it; switch to a bypassing agent (FEIBA or rFVIIa). Reflexively escalating factor dose for a high-responder is wrong.

5. ITI — immune tolerance

WHAT YOU SEE

A center 'ITI CHAMBER' (immune tolerance induction) — repeated factor dosing to desensitize and eradicate the inhibitor

WHAT IT ENCODES

Immune tolerance induction (ITI) is the strategy to ERADICATE an inhibitor: regular, often high-dose factor exposure over months to years to desensitize the immune system and restore normal factor responsiveness. It is the definitive long-term fix, with success rates around 60–80% in hemophilia A.

BOARD TRAP

ITI is the long-term eradication strategy, NOT an acute-bleed treatment — during an active bleed in an inhibitor patient you reach for a bypassing agent, not ITI; confusing the two is a tested distinction.

6. Emicizumab bridges IXa-X

WHAT YOU SEE

A center emicizumab panel with a Y-shaped antibody 'mimics FVIII — no FVIII needed' — the bispecific that bridges FIXa to FX

WHAT IT ENCODES

Emicizumab is a bispecific monoclonal antibody that simultaneously binds activated factor IX (FIXa) and factor X, physically bridging them to mimic the cofactor function of FVIII. Because it is not factor VIII, it works even in the presence of FVIII inhibitors and is given subcutaneously weekly for PROPHYLAXIS — transforming care for hemophilia A with inhibitors.

BOARD TRAP

Emicizumab is for hemophilia A PROPHYLAXIS (with or without inhibitors); it mimics FVIII so it does NOT help hemophilia B (which lacks FIX, not VIII) — and it is prophylaxis, not acute-bleed rescue. Using it for B, or as on-demand bleed treatment, is wrong.

7. Emicizumab + aPCC danger

WHAT YOU SEE

Bottom-left 'emicizumab + aPCC = fatal thrombotic microangiopathy, do NOT co-administer'

WHAT IT ENCODES

Patients on emicizumab who develop a breakthrough bleed and receive activated PCC (FEIBA) are at risk of THROMBOTIC MICROANGIOPATHY and serious thrombosis — a potentially fatal interaction, especially with cumulative/high-dose aPCC. Prefer rFVIIa at the lowest effective dose for breakthrough bleeds on emicizumab.

BOARD TRAP

Do NOT combine emicizumab with aPCC (FEIBA) — the combination caused fatal TMA/thrombosis in trials; for a breakthrough bleed on emicizumab, low-dose rFVIIa is the safer bypassing agent.

8. aPTT unreliable on emicizumab

WHAT YOU SEE

A center note 'standard aPTT unreliable on emicizumab — use chromogenic FVIII' — the drug falsely shortens aPTT-based assays

WHAT IT ENCODES

Emicizumab activates FIX/X directly and so dramatically shortens/normalizes the standard aPTT, making aPTT-based assays (including one-stage FVIII activity and Bethesda titers) falsely reassuring/unreliable. Monitor with a BOVINE-based chromogenic FVIII assay, which is insensitive to emicizumab and reflects true FVIII/inhibitor status.

BOARD TRAP

Do NOT trust a normal aPTT to gauge hemostasis or factor level in a patient on emicizumab — it is artificially shortened; only a bovine chromogenic assay gives valid FVIII activity and inhibitor measurement.

9. FXI deficiency (Hemophilia C)

WHAT YOU SEE

The right alcove 'FXI deficiency' with Ashkenazi/Iraqi-Jewish figures — autosomal hemophilia C with unpredictable bleeding

WHAT IT ENCODES

Factor XI deficiency (hemophilia C) is autosomal (not X-linked), strongly enriched in Ashkenazi and Iraqi Jews. Bleeding is mild and UNPREDICTABLE — typically injury/surgery-related, especially at high-fibrinolytic mucosal sites (oral cavity, nasopharynx, urinary/genital tract), rather than spontaneous hemarthrosis. It prolongs the aPTT.

BOARD TRAP

In FXI deficiency, bleeding severity does NOT correlate with the factor level or the degree of aPTT prolongation — a very low level may not bleed and a near-normal level may; management is guided by personal/family bleeding history, not the number.

10. FXI bleeding is variable

WHAT YOU SEE

FXI note 'clean surgical history safe, but excise high-fibrinolysis tissue carefully'

WHAT IT ENCODES

Because FXI bleeding is so unpredictable, the practical approach is risk-stratify by HISTORY and surgical SITE: a patient with a clean prior surgical/dental history is generally safe, but procedures on high-fibrinolytic tissue (tonsils, dental, urologic) warrant prophylaxis (antifibrinolytics like tranexamic acid ± FFP/FXI concentrate).

BOARD TRAP

Do NOT give prophylactic factor to an asymptomatic FXI-deficient patient based on a prolonged aPTT alone — treat according to bleeding history and the bleeding risk of the specific procedure/site, not the lab value.

11. FXII/HMWK/prekallikrein

WHAT YOU SEE

A right '#2' note 'FXII / HMWK / prekallikrein — prolonged aPTT, ZERO bleeding, do NOT treat' — harmless contact-factor deficiencies

WHAT IT ENCODES

Deficiencies of the contact-pathway factors — factor XII (Hageman), high-molecular-weight kininogen (HMWK), and prekallikrein — markedly PROLONG the aPTT but cause NO clinical bleeding (they aren't needed for in-vivo hemostasis; some FXII deficiency is even prothrombotic). They are typically incidental lab findings.

BOARD TRAP

A long aPTT from FXII/HMWK/prekallikrein deficiency is HARMLESS — do NOT treat it and do NOT delay or cancel surgery for it; transfusing or working it up as a bleeding risk is the classic wrong answer.

12. FXIII — normal PT & aPTT

WHAT YOU SEE

Bottom-right 'FXIII deficiency — all standard tests normal, diagnosis missed' panel — clot-stabilizer defect invisible on PT/aPTT

WHAT IT ENCODES

Factor XIII cross-links fibrin to stabilize the clot, so its deficiency causes DELAYED bleeding after initial hemostasis, poor wound healing, recurrent miscarriage, and the classic delayed umbilical-stump bleeding in neonates. Crucially, PT, aPTT, fibrinogen and platelet count are ALL NORMAL — diagnosis requires a urea/clot-solubility assay or FXIII level.

BOARD TRAP

FXIII deficiency is MISSED on routine coagulation testing because PT, aPTT and platelets are all NORMAL — suspect it with delayed bleeding, umbilical-stump bleeding, poor healing or recurrent pregnancy loss despite a normal coag panel, and order the clot-solubility (urea) test.

13. Anaphylaxis risk in FIX

WHAT YOU SEE

A left gauge 'FIX inhibitor -> anaphylactic reactions' — the signature hazard unique to factor IX inhibitors

WHAT IT ENCODES

Factor IX inhibitors are much rarer than FVIII inhibitors (~3–5% of severe hemophilia B) but carry distinctive hazards: severe ANAPHYLACTIC/allergic reactions on FIX exposure and an association with nephrotic syndrome (notably during ITI). They occur particularly in patients with large gene deletions/null FIX mutations.

BOARD TRAP

ANAPHYLAXIS is the signature danger of FIX inhibitors (with nephrotic syndrome during ITI) — it is NOT a feature of FVIII inhibitors; early FIX infusions in a high-risk B patient should be given where anaphylaxis can be managed.